LAB 7 – DNS Configuration and Troubleshooting

Curso: *The Bits and Bytes of Computer Networking*
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Año: *15/04/2026*

1. Introduction

This lab aims to identify and troubleshoot failures in DNS services within a local network using Cisco Packet Tracer.

The lab simulates real-world scenarios where DNS misconfiguration affects connectivity. Through testing and analysis, the goal is to understand how to diagnose and resolve these issues.

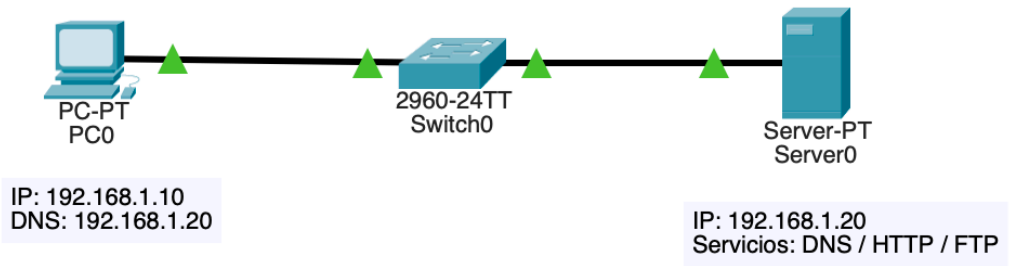
2. Network Topology

The network used in this lab consists of:

- One PC acting as a client
- One Cisco 2960 switch
- One server providing DNS services

All devices are connected within the same LAN.

Figure 1 – Network topology



3. Network Configuration

Static IP addresses were configured as follows:

Device	IP Address	DNS Server
PC0	192.168.1.10	192.168.1.20
Server	192.168.1.20	—

Figure 2 – PC IP configuration

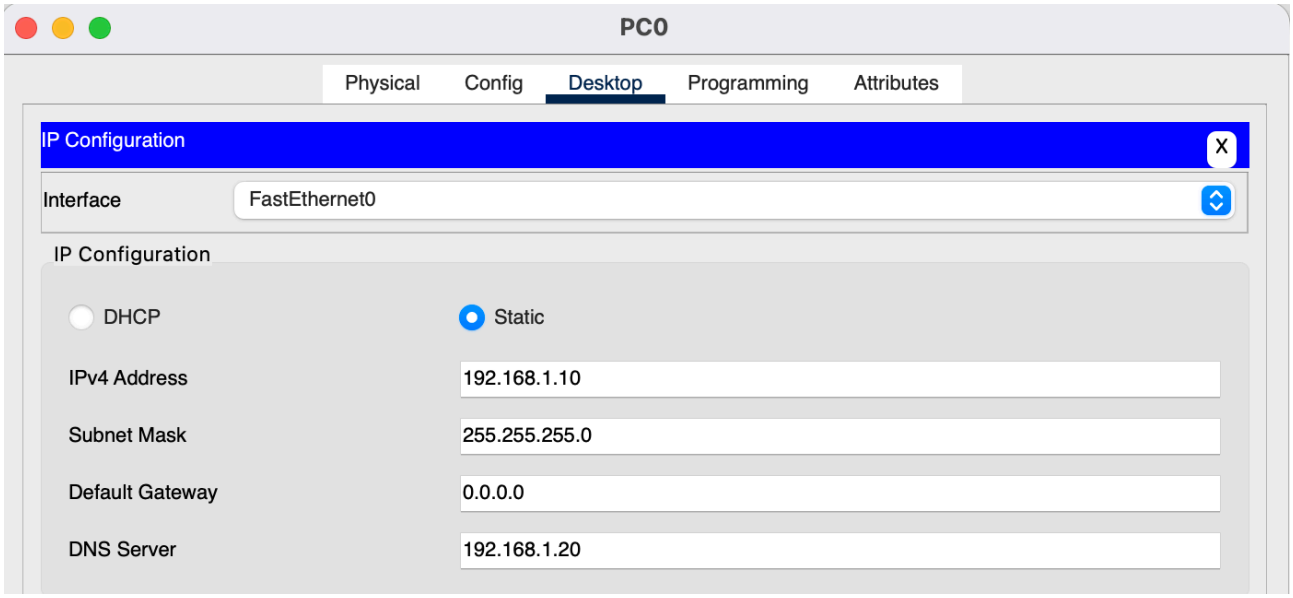
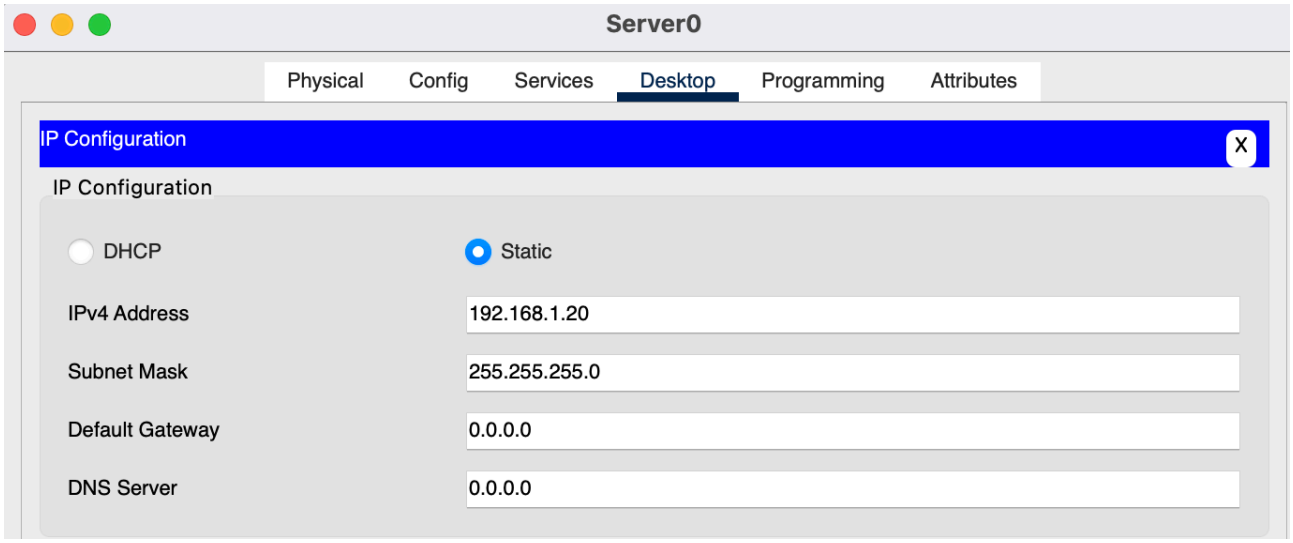


Figure 3 – Server IP configuration

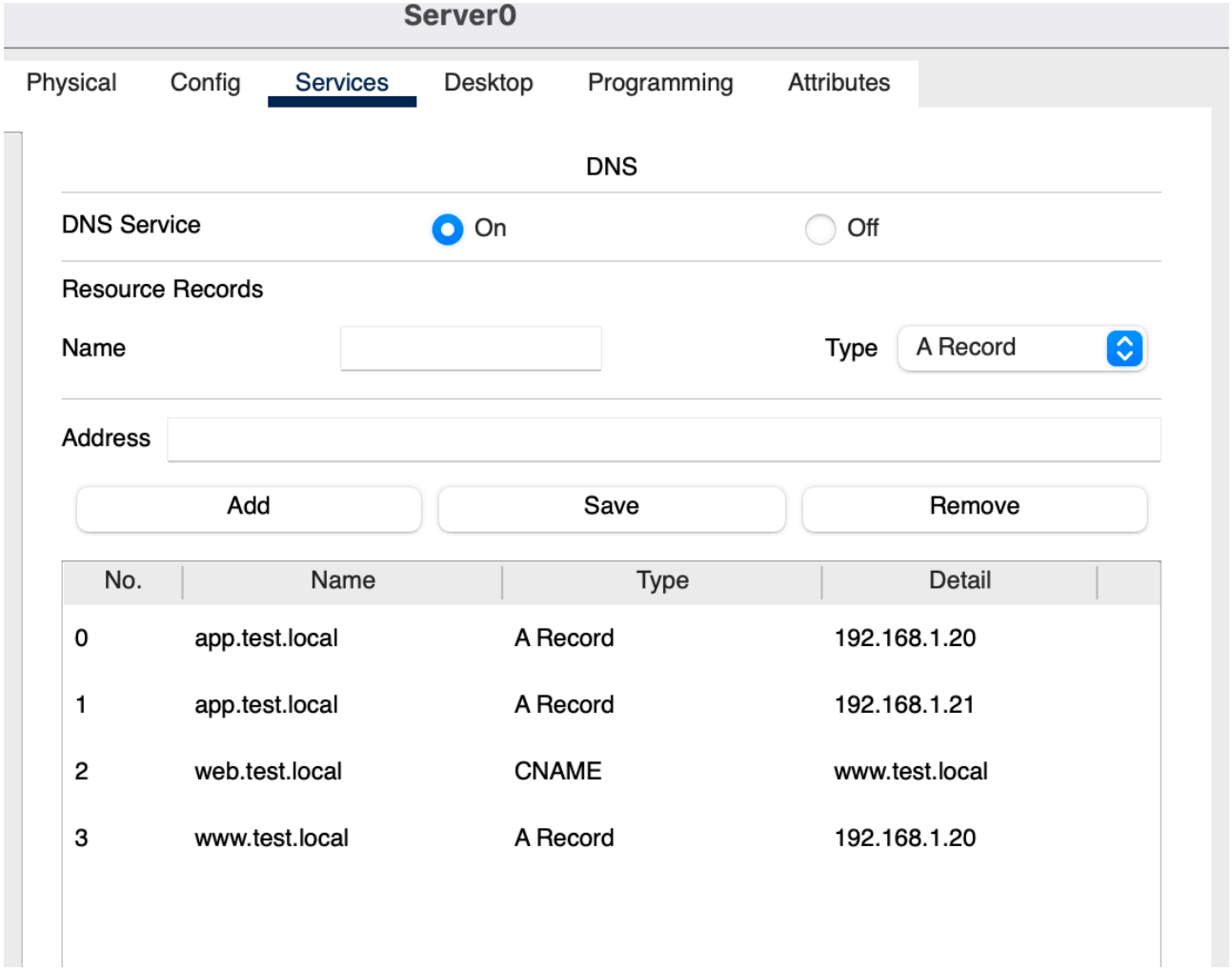


4. DNS Configuration

The server was configured with the following DNS records:

- **A Record**
  www.test.local → 192.168.1.20
- **CNAME Record**
  web.test.local → www.test.local
- **Multiple A Records (Round Robin concept)**
  app.test.local → 192.168.1.20
  app.test.local → 192.168.1.21

Figure 4 – DNS records configuration (A and CNAME)



5. Verification

Initial tests were performed to verify that DNS resolution was working correctly.

ping www.test.local

The domain resolved successfully to 192.168.1.20.

Figure 5 – Successful DNS resolution

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping www.test.local

Pinging 192.168.1.20 with 32 bytes of data:

Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

6. CNAME Verification

The configured alias was tested:

ping web.test.local

The domain web.test.local resolved correctly to www.test.local, returning the same IP address.

Figure 6 – CNAME resolution via ping

```
C:\>ping web.test.local

Pinging 192.168.1.20 with 32 bytes of data:

Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time=9ms TTL=128

Ping statistics for 192.168.1.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 9ms, Average = 2ms
```

7. DNS Failure Scenario

7.1 Problem Description

The client PC was unable to access the domain www.test.local.

7.2 Diagnosis

The following tests were performed:

ping www.test.local → fails
ping 192.168.1.20 → succeeds

This indicates that network connectivity is working, but name resolution is failing, suggesting a DNS configuration issue.

Figure 7 – Failed domain resolution

```
C:\>ping www.test.local

Pinging 192.168.1.99 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.99:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Figure 8 – Successful ping to IP address

```
C:\>ping 192.168.1.20

Pinging 192.168.1.20 with 32 bytes of data:

Reply from 192.168.1.20: bytes=32 time=11ms TTL=128
Reply from 192.168.1.20: bytes=32 time=1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 3ms
```

7.3 Analysis

The DNS server configured on the client was incorrect (192.168.1.99).

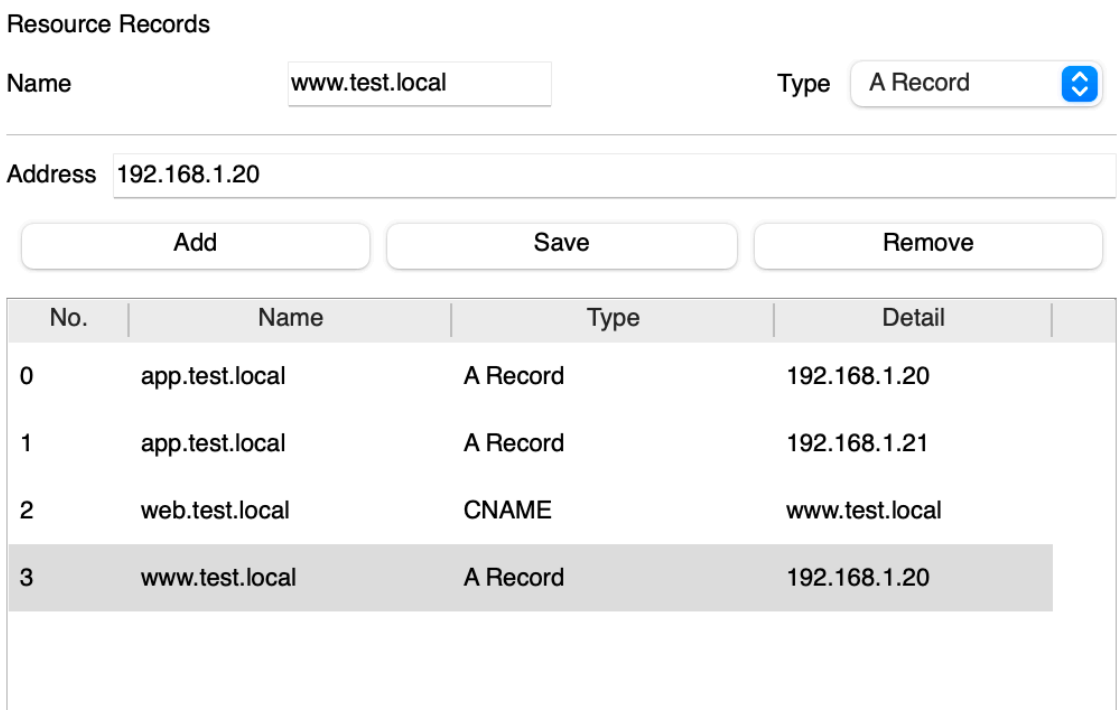
As a result, the domain was resolved incorrectly or failed to resolve, preventing proper communication.

However, direct IP connectivity remained functional, confirming that the issue was related to DNS and not the network itself.

7.4 Solution

The DNS configuration on the client PC was corrected by setting the proper DNS server address (192.168.1.20).

Figure 9 – Correct DNS configuration on client



After applying the fix, the domain resolved correctly.

ping www.test.local → successful replies

Figure 10 – Successful domain resolution

```
C:\>ping www.test.local

Pinging 192.168.1.20 with 32 bytes of data:

Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

8. Conclusion

This lab demonstrated how DNS operates in a local network and how to troubleshoot common DNS issues.

Key takeaways:

- DNS translates domain names into IP addresses
- Incorrect DNS configuration can break network services
- Testing both domain names and IP addresses helps isolate issues
- Troubleshooting requires identifying whether the issue is DNS or network-related